

I3DION Spatial

Product Requirements Document (PRD)

Vision, Problem Statement, Solution, Target Users, MVP Scope & Success Metrics

Vision

Industrial sales and product visualization have remained largely dependent on static catalogs, brochures, images, and product videos for decades. While these methods provide information, they often fail to communicate how a product will look, fit, operate, or interact within a real-world environment.

I3DION Spatial is envisioned as a next-generation Industrial AR Catalog and Sales Visualization Platform that transforms traditional product communication into immersive, interactive experiences. The platform enables manufacturers, dealers, and sales teams to present products digitally in a way that allows customers to understand, explore, and evaluate products before making purchasing decisions.

The long-term vision of I3DION Spatial is to become the standard digital visualization platform for industrial products by combining interactive catalogs, 3D visualization, QR-powered experiences, and spatial computing technologies into a single ecosystem.

Rather than replacing existing sales processes, the platform enhances them by providing a more engaging and informative method of product presentation.

Problem Statement

Industrial product sales involve unique challenges that are not effectively addressed by conventional marketing and sales materials.

Large industrial products such as compressors, pumps, generators, manufacturing equipment, textile machinery, process equipment, and heavy engineering systems are difficult to demonstrate physically due to their size, weight, transportation costs, and installation requirements.

Most companies currently rely on PDF catalogs, brochures, product images, videos, and sales presentations. While these materials provide technical information, they do not help customers visualize products in their actual environments.

As a result, customers frequently face uncertainty regarding:

- Product dimensions and scale

- Installation feasibility
- Space requirements
- Maintenance accessibility
- Product appearance in real environments
- Operational understanding

For sales teams, the challenge is even greater. Demonstrating large industrial products often requires extensive explanations, site visits, repeated meetings, and lengthy communication cycles.

This gap between product information and customer understanding can slow purchasing decisions, reduce engagement, and increase the effort required to close sales opportunities.

There is a clear need for a solution that enables industrial products to be visualized, explored, and experienced without requiring physical demonstrations.

Solution

I3DION Spatial addresses these challenges by converting traditional product catalogs into interactive digital experiences powered by 3D visualization, QR technology, and Augmented Reality.

The platform allows companies to upload product information, images, videos, and 3D models to create interactive digital catalogs that can be shared with customers through web links or QR codes.

When a customer scans a QR code, the product experience opens directly in a mobile browser, eliminating the need for application installation.

Customers can:

- Explore products in 3D
- Place products in real-world environments using AR
- View product specifications
- Interact with product hotspots
- Watch operational animations
- Access technical documents
- Contact sales teams directly

One of the platform's key capabilities is spatial product visualization. Users can place products within their environment, lock product positions, and physically move around the product to inspect it from different perspectives.

This creates a highly immersive product experience that improves customer understanding while reducing the dependency on physical demonstrations.

Ultimately, I3DION Spatial serves as a bridge between industrial products and digital experiences, enabling more effective communication, faster evaluation, and better purchasing decisions.

Target Users

Primary Users

Industrial Manufacturers

Manufacturers of compressors, pumps, generators, machinery, automation systems, process equipment, and engineering products who require modern methods to showcase and demonstrate their products.

Industrial Dealers and Distributors

Organizations responsible for selling multiple products and brands that need a centralized visualization platform to improve customer engagement.

Sales Engineers

Field sales professionals who require effective tools to explain products, demonstrate capabilities, and communicate value propositions during customer interactions.

Product Marketing Teams

Teams responsible for maintaining product information, creating catalogs, managing product assets, and supporting sales activities.

Secondary Users

Architects and Consultants

Professionals involved in planning and evaluating equipment placement within facilities and projects.

Facility Managers

Individuals responsible for evaluating installation feasibility and operational requirements.

End Customers

Decision-makers who need a better understanding of products before purchasing.

MVP Scope (Version 1.0)

The first version of I3DION Spatial will focus on validating the core product vision and delivering the primary value proposition.

Company Management

- Company profile creation
- Brand identity configuration
- Logo management
- Contact information management

Product Management

- Product creation and editing
- Product image uploads
- Product video uploads
- GLB model uploads
- Product specification management

Catalog Builder

- Catalog creation
- Product selection
- Catalog publishing
- Digital catalog management

QR Experience Engine

- Product QR generation
- Catalog QR generation
- QR download and sharing

Product Experience Page

- Product overview
- Product specifications
- Product media gallery
- Product documents

Interactive 3D Viewer

- Product rotation
- Product zoom
- Product inspection

AR Visualization

- Product placement
- Product repositioning
- Product scaling
- Product rotation

Product Lock System

- Lock product position
- Unlock product position
- Reset product placement

Animation System

- Play product animations
- Pause animations
- Demonstration sequences

Lead Generation

- Request quotation
- Contact sales
- Inquiry submission

Success Metrics

The success of I3DION Spatial will be measured across product adoption, user engagement, and business impact.

Product Development Metrics

MVP Completion

All planned Version 1.0 features delivered within the target development timeline.

Platform Stability

Successful operation across supported browsers and devices without critical issues.

Engagement Metrics

Catalog Views

Measure the number of catalog visits generated through shared links and QR codes.

QR Scans

Track the number of successful QR interactions.

Product Experience Sessions

Monitor the number of users interacting with product pages and AR experiences.

Session Duration

Measure the average time users spend exploring products.

Product Interaction Rate

Track interactions including animations, hotspots, AR placement, and product exploration.

Business Metrics

Company Adoption

Number of companies actively using the platform.

Product Uploads

Number of products published on the platform.

Lead Generation

Number of customer inquiries generated through the platform.

Sales Enablement Impact

Qualitative feedback from sales teams regarding improved product demonstrations and customer understanding.

Strategic Outcome

The primary objective of I3DION Spatial Version 1.0 is not to build a generic AR viewer. The objective is to establish a scalable Industrial AR Catalog and Sales Visualization Platform that modernizes how industrial products are presented, understood, and evaluated.

By focusing on solving real-world sales and visualization challenges, I3DION Spatial aims to create a foundation for future capabilities including advanced product configurators, intelligent recommendations, space validation, analytics, and enterprise collaboration tools.